



S. B. JAIN INSTITUTE OF TECHNOLOGY, MANAGEMENT & RESEARCH, NAGPUR.

Practical No. 2

AIM: A college placement cell requires every student to create a professional online profile for recruiters. Design a semantic HTML webpage that displays your personal information, photograph, education details, hobbies, and contact information.

NAME OF STUDENT	
USN NO.	
DATE OF PERFORMANCE	
DATE OF SUBMISSION	

OBSERVATION (Practical No. 2)

Application Software:	
System Software (OS):	
Web Application URL:	

OBSERVATIONS:

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.

Regular Performance in Lab (3)	
Faculty Signature & Date	

AIM: A college placement cell requires every student to create a professional online profile for recruiters. Design a semantic HTML webpage that displays your personal information, photograph, education details, hobbies, and contact information.

OBJECTIVES

- To understand the importance of semantic HTML tags in modern web accessibility and SEO engine discovery layouts.
- To structure a clean, human-readable professional placement portfolio using structured semantic sectioning elements.
- To properly integrate embedded multimedia objects () and structural data arrays (<table>,) within web text components.

THEORY

1. Introduction to Semantic HTML5 Architecture

Semantic HTML refers to the engineering practice of utilizing HTML tags that explicitly describe the meaning and purpose of their enclosed content to both the browser and the developer, rather than using non-semantic structural wrappers like <div> or .

Prior to HTML5, developers relied heavily on complex nested division layouts distinguished only by custom class or ID tags (e.g., <div class="header">). HTML5 introduced clear, dedicated semantic container elements that create a standardized document structure:

- <header>: Identifies the introductory content, layout titles, or primary branding tags of a webpage or structural section.
- <nav>: Defines a block of navigation links intended for site or page-level orientation controls.
- <main>: Encapsulates the unique, central core functional content of the specific document body. There must only be one visible <main> element per document.
- <section>: Represents a standalone structural segment of a page that shares a single thematic topic (e.g., Education, Projects).
- <article>: Outlines a self-contained compilation of text intended to be independently reusable or distributable.
- <aside>: Wraps secondary or tangential content loosely related to the surrounding page modules (e.g., a list of secondary skill metrics or hobby tags).
- <footer>: Marks the concluding baseline elements of a page, typically aggregating copyright notices, legal disclaimers, or metadata links.

2. Accessibility, Document Object Model (DOM) Parsing, and SEO

Deploying semantic markup is a requirement for production-grade software engineering for several reasons:

- **Accessibility (A11y):** Screen readers and assistive technologies parse the DOM using semantic milestones. For example, a visually impaired user can jump directly to the `<main>` section or skip across distinct `<section>` nodes dynamically.
- **Search Engine Optimization (SEO):** Web crawling algorithms evaluate semantic tags to decipher page hierarchy. Headings (`<h1>` through `<h6>`) and structurally relevant sections pass cleaner metadata signals to indexing algorithms, boosting page rankings.
- **Code Maintainability:** Standardized semantic scoping reduces code layout fragmentation, allowing new engineering teams to audit and style the interface quickly.

CSS CLASS TABLE

#	Tailwind Class	Meaning	Purpose
1	bg-gradient-to-r	Background gradient from left to right	Creates a colorful gradient background
2	from-indigo-500	Start the gradient with indigo color	Sets the starting color
3	via-purple-500	Middle color of the gradient	Creates a smooth color transition
4	to-pink-500	End the gradient with pink color	Sets the ending color
5	min-h-screen	Minimum height = full screen	Makes the page occupy the full browser height
6	py-10	Vertical padding (top & bottom)	Adds space above and below the content
7	text-center	Center-align text	Centers headings and paragraphs
8	text-white	White text color	Makes text white
9	mb-8	Bottom margin	Creates space below the header
10	text-4xl	Extra-large text	Makes the heading large
11	font-bold	Bold font	Makes text bold
12	max-w-3xl	Maximum width = 48rem (768px)	Prevents content from becoming too wide
13	mx-auto	Auto left & right margins	Centers the container horizontally
14	bg-white	White background	Gives the profile card a white background

Web Fundamentals and Basic Frontend Design Lab (N-MDMWD301P)

15	rounded-xl	Extra-large rounded corners	Makes the card corners rounded
16	shadow-2xl	Large shadow	Gives a floating card effect
#	Tailwind Class	Meaning	Purpose
17	overflow-hidden	Hide overflowing content	Prevents content from spilling outside the card
18	bg-purple-700	Dark purple background	Styles the profile section
19	py-8	Vertical padding	Adds space inside the profile section
20	w-36	Width = 9rem (144px)	Sets the image width
21	h-36	Height = 9rem (144px)	Sets the image height
22	rounded-full	Fully rounded corners	Makes the image circular
23	border-4	4px border	Adds a border around the image
24	border-white	White border color	Makes the border white
25	text-3xl	Large text	Styles the student's name
26	mt-4	Top margin	Adds space above the name
27	text-purple-100	Light purple text	Styles the subtitle
28	p-6	Padding on all sides	Adds space inside sections
29	border-b	Bottom border	Separates sections
30	text-2xl	Large heading	Styles section titles
31	text-purple-700	Dark purple text	Colors section headings
32	mb-4	Bottom margin	Adds space below headings
33	space-y-2	Vertical gap between child elements	Adds spacing between paragraphs
34	bg-purple-50	Very light purple background	Highlights alternate sections
35	w-full	Width = 100%	Makes the table fill the container
36	border	Adds a border	Creates table borders
37	border-gray-300	Light gray border	Colors table borders
38	bg-purple-200	Light purple background	Styles the table header
39	p-2	Padding	Adds space inside table cells
40	list-disc	Bulleted list	Displays bullets
41	list-inside	Bullets inside the content	Aligns bullets within the list
42	space-y-1	Small vertical gap	Adds spacing between list items
43	py-6	Vertical padding	Adds space inside the footer

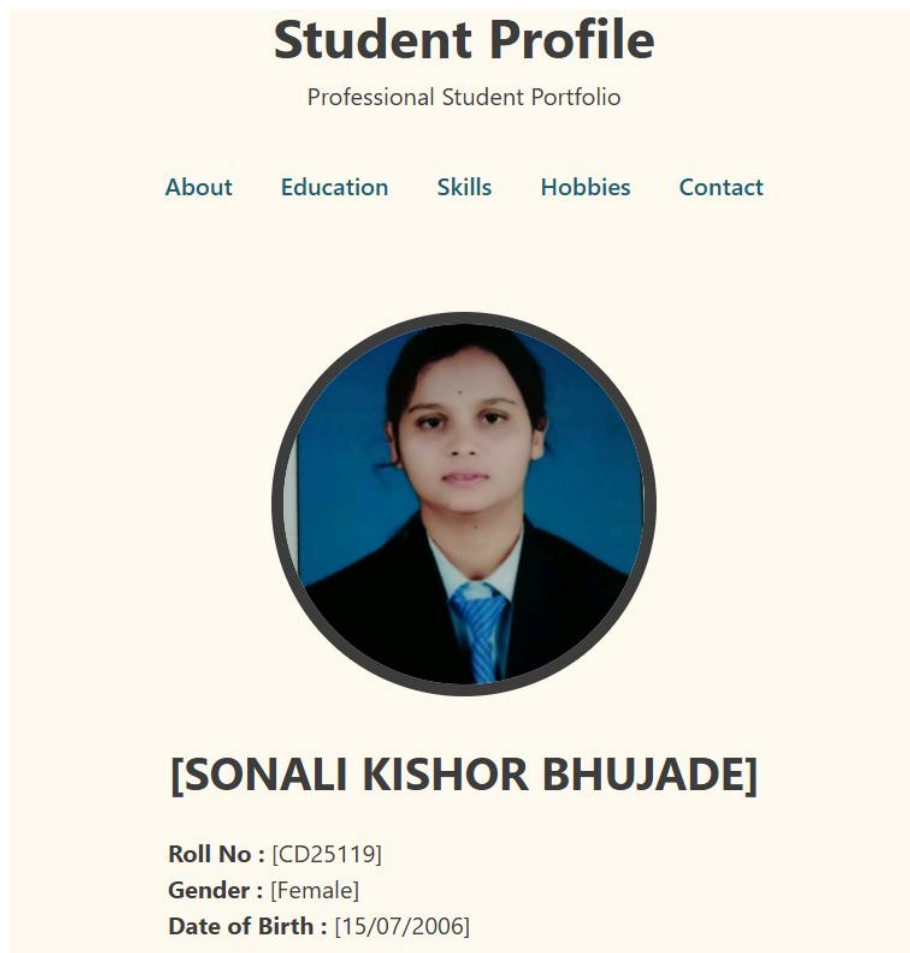
SOURCE CODE GITHUB REPOSITORY

<https://github.com/sonalibds25-creator>

SOURCE CODE GITHUB REPOSITORY QR CODE



OUTPUT SCREENSHOTS



Technical Skills



Hobbies

- Reading Technical Books
- Programming
- Listening to Music
- Playing Cricket
- Learning New Technologies

Career Objective

To become a skilled Software Engineer by continuously improving my programming, problem-solving, and communication skills while contributing effectively to an innovative organization.

Contact Information

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REVIEW QUESTIONS

1. What makes an HTML element "semantic," and how does it differ from structural styling wrappers like `<div>`?

2. Why is the `<main>` element structurally constrained to appear only once as a visible root element within a page document?

3. What is the structural purpose of the `alt` attribute within an `<img>` tag, and why is it legally mandated for accessibility conformance?

4. How do search engine optimization (SEO) crawlers interpret heading elements (`<h1>` through `<h6>`) to determine content hierarchy?

5. What is the primary semantic difference between an unordered list () and an ordered list () container?

CONCLUSION

REFERENCES

- <https://developer.mozilla.org/en-US/docs/Glossary/Semantics>
- <https://www.w3.org/TR/html52/>
- <https://www.w3.org/WAI/fundamentals/accessibility-intro/>

EVALUATION

Regular Performance in Lab (3)	
Viva-Voce (3)	
Quality of Submission (4)	
Faculty Signature & Date	